Light

GCSE Computer Science – Networks Basics (Set 3)

Over **1,100,000+ marks** of exam-aligned content across **67 qualifications**. 4,200+ auto-marked assessments & long-answer practice. No login required.

GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Networks Basics

Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Which of the following is required for a collection of computers to be considered a network?

- ○ **A.** They must be connected together and able to communicate
- ○ **B.** They must all be the same brand
- ○ **C.** They must all run the same operating system
- ○ **D.** They must be located in different countries

Q2 (1 mark)

A museum has installed a network connecting all the computers in its building. This network would be classified as which type?

- ○ **A.** A Wide Area Network (WAN)
- ○ **B.** A Personal Area Network (PAN)
- ○ **C.** A Bluetooth network
- ○ **D.** A Local Area Network (LAN)

Q3 (1 mark)

Which characteristic best identifies a Wide Area Network (WAN)?

- ○ **A.** It covers a wide geographical area, potentially spanning cities or countries
- ○ **B.** It can only use wireless connections
- ○ **C.** It must be owned by a single person
- ○ **D.** It connects fewer than ten computers

Q4 (1 mark)

A fitness tracker syncing data with a smartphone via Bluetooth is an example of which type of network?

- ○ **A.** A Wide Area Network (WAN)
- ○ **B.** A Local Area Network (LAN)
- ○ **C.** The internet
- ○ **D.** A Personal Area Network (PAN)

Q5 (1 mark)

A veterinary practice connects all its computers to a network. Which is an advantage of this setup?

- ○ **A.** Each computer will need its own separate internet connection
- ○ **B.** Staff will not be able to communicate electronically
- ○ **C.** Patient records can be accessed from any computer on the network
- ○ **D.** Each computer will require more storage space

Q6 (1 mark)

What is the main purpose of the Internet Protocol (IP)?

- ○ **A.** To encrypt all data before transmission
- ○ **B.** To display web pages in a browser
- ○ **C.** To send emails between users
- ○ **D.** To address and route data packets across networks

Q7 (1 mark)

What makes TCP a 'reliable' protocol?

- ○ **A.** It ensures data packets arrive complete and in the correct order
- ○ **B.** It makes internet connections faster
- ○ **C.** It blocks all viruses and malware

- **D.** It only works with wired connections

Q8 (1 mark)

When you type a website address into a browser, which protocol is used to request the web page from the server?

- **A.** SMTP
- **B.** IMAP
- **C.** HTTP or HTTPS
- **D.** Bluetooth

Q9 (1 mark)

What type of network security device monitors traffic and blocks connections based on security rules?

- **A.** A monitor
- **B.** A keyboard
- **C.** A printer
- **D.** A firewall

Q10 (1 mark)

Why are wireless networks generally considered less secure than wired networks?

- **A.** Wireless signals travel through the air and could potentially be

intercepted

- ○ **B.** Wireless networks are always slower
- ○ **C.** Wireless networks cannot use encryption
- ○ **D.** Wireless networks use more electricity

Q11 (1 mark)

Which statement about copper network cables is correct?

- ○ **A.** Copper cables are immune to electrical interference
- ○ **B.** Copper cables transmit data using light pulses
- ○ **C.** Copper cables are generally less expensive than fibre optic cables
- ○ **D.** Copper cables can only be used for wireless networks

Q12 (1 mark)

At which layer of the TCP/IP model does the IP protocol operate?

- ○ **A.** Application layer
- ○ **B.** Transport layer
- ○ **C.** Internet layer
- ○ **D.** Link layer

Q13 (2 marks)

A small theatre is considering installing a network to connect its box office, backstage computers, and administrative office. What are potential disadvantages they should consider?

- ○ **A.** Initial setup costs for equipment, potential security vulnerabilities, and need for technical expertise to maintain the network
- ○ **B.** Staff will no longer be able to print documents
- ○ **C.** Computers will only work when connected to the internet
- ○ **D.** Files will no longer be able to be saved on any computer

Q14 (2 marks)

A recording studio requires a network for transferring large audio files between workstations. Why might wired networking be preferred over wireless in this environment?

- ○ **A.** Wireless cannot transfer audio files
- ○ **B.** Wired connections provide faster, more consistent speeds and are not affected by interference from audio equipment
- ○ **C.** Wired networks do not require any equipment
- ○ **D.** Wireless networks do not work indoors

Q15 (2 marks)

A film studio needs to transfer 4K video files (often 100GB or larger) between editing suites located in different buildings across their campus. Why would fibre optic cables be the best choice?

- ○ **A.** Fibre optic is the cheapest option available

- ○ **B.** Fibre optic provides the highest bandwidth for large file transfers and can span longer distances without signal degradation
- ○ **C.** Fibre optic cables require no installation
- ○ **D.** Fibre optic only works for video files, not other data types

Q16 (2 marks)

An employee sends an email from their work computer and later reads the reply on their tablet at home. Describe how SMTP and IMAP are used in this process.

- ○ **A.** SMTP displays the email on screen, IMAP encrypts it for security
- ○ **B.** SMTP and IMAP both send emails; they are interchangeable
- ○ **C.** SMTP transfers the outgoing email to the server; IMAP allows the tablet to retrieve and access the reply from the server
- ○ **D.** IMAP sends the email first, then SMTP receives the reply

Q17 (2 marks)

A gym wants members to log in before accessing the gym's Wi-Fi network. What security method is this, and how does it help protect the network?

- ○ **A.** Encryption—it scrambles all data on the network
- ○ **B.** MAC filtering—it checks the type of device being used
- ○ **C.** Authentication—it verifies user identity before granting network access, ensuring only registered members can connect

- ○ **D.** Firewall—it monitors all gym equipment

Q18 (2 marks)

A law firm transmits confidential client documents across its network. Why is encryption essential for this data?

- ○ **A.** Encryption makes documents download faster
- ○ **B.** Encryption automatically backs up all documents
- ○ **C.** Encryption prevents employees from printing documents
- ○ **D.** Encryption ensures only authorised recipients can read the documents, even if the data is intercepted during transmission

Q19 (2 marks)

A hotel uses MAC address filtering on its conference room network. When a guest's laptop cannot connect despite entering the correct password, what is the most likely cause?

- ○ **A.** The laptop's battery is too low
- ○ **B.** The laptop's MAC address has not been added to the approved list on the network
- ○ **C.** The laptop is too old to connect to any network
- ○ **D.** MAC filtering only works with desktop computers

Q20 (2 marks)

SMTP, IMAP, HTTP, and HTTPS all operate at the application layer of the TCP/IP model. What characteristic do these protocols share that

explains this classification?

- ○ **A.** They all handle physical network cable connections
- ○ **B.** They all assign IP addresses to devices
- ○ **C.** They all directly support user-facing applications like email programs and web browsers
- ○ **D.** They all break data into packets for transmission

Q21 (2 marks)

When a computer sends data across a network, what happens at the link layer of the TCP/IP model?

- ○ **A.** The network interface card (NIC) transmits the data as physical signals through the network medium
- ○ **B.** The web browser formats the data for display
- ○ **C.** Email protocols sort messages into folders
- ○ **D.** The data is given an IP address for routing

Q22 (2 marks)

A website for a charity accepting donations changes from HTTP to HTTPS. What protection does this provide for donors?

- ○ **A.** HTTPS guarantees that the charity is legitimate and trustworthy
- ○ **B.** HTTPS encrypts donation details (credit card numbers, personal information) during transmission, protecting them from interception
- ○ **C.** HTTPS automatically donates extra money to the charity

○ **D.** HTTPS prevents the website from ever going offline

Q23 (2 marks)

A company's firewall is configured to block outgoing connections to social media websites during work hours. What does this demonstrate about firewall capabilities?

○ **A.** Firewalls can only block incoming connections, so this would not work

○ **B.** Firewalls only work during specific hours of the day

○ **C.** Firewalls automatically know which websites are social media

○ **D.** Firewalls can monitor and control both incoming and outgoing network traffic based on defined rules

Q24 (3 marks)

A user at a farm sends an email containing crop reports to their agricultural supplier in another county. Describe how the email data flows through the four layers of the TCP/IP model when being sent.

○ **A.** Link → Internet → Transport → Application: physical signals are sent first, then the email is written

○ **B.** Application → Transport → Internet → Link: SMTP prepares the email, TCP segments it into packets, IP adds addressing and routing, the NIC transmits physical signals

○ **C.** Transport → Application → Link → Internet: TCP processes data before the email is composed

○ **D.** Internet → Link → Application → Transport: IP addressing happens before the email application is used

Q25 (3 marks)

A cruise ship has passengers using Bluetooth speakers with their phones, a shipboard network connecting entertainment systems and point-of-sale terminals, and satellite internet connecting to shore-based reservation systems. Identify the three network types involved.

- ○ **A.** Three LANs: all networks are on the same ship
- ○ **B.** Three WANs: all connections eventually reach the shore
- ○ **C.** PAN (Bluetooth speaker to phone), LAN (shipboard entertainment and POS network), WAN (satellite connection to shore-based systems)
- ○ **D.** Three PANs: all connections involve passengers' personal devices

Q26 (3 marks)

A medical laboratory handles sensitive test results that must be transmitted securely. The IT team proposes implementing a firewall, encryption, and authentication together. Evaluate why this combined approach is more effective than using any single method alone.

- ○ **A.** Using all three is wasteful because each method provides identical protection
- ○ **B.** The firewall blocks external attacks, encryption protects data content if intercepted, and authentication verifies user identity—each addresses different threats, creating multiple barriers an attacker must overcome
- ○ **C.** Only encryption is needed for medical data; the other methods are unnecessary
- ○ **D.** These methods conflict with each other and should not be used together

Q27 (3 marks)

A new restaurant is planning its network. The kitchen has fixed workstations for order management, the dining area needs tablets for servers to take orders, and customers expect free Wi-Fi. Design the most appropriate network configuration.

- ○ **A.** Use only wired connections throughout—wireless is too unreliable for a business
- ○ **B.** Use only wireless for everything—it is easier to install
- ○ **C.** Use wired connections for fixed kitchen workstations (reliable, consistent), wireless for server tablets (mobility needed) and customer Wi-Fi (convenience)
- ○ **D.** Use Bluetooth for all connections since the range is sufficient for a restaurant

Q28 (3 marks)

A new sports stadium is being built. The network will connect 100 point-of-sale terminals, digital signage systems, and 50 CCTV cameras across a large site with connections between buildings up to 300 metres apart. Some areas are near high-powered lighting equipment. Evaluate the cable types needed.

- ○ **A.** Copper cables throughout—they are cheaper and sufficient for all needs
- ○ **B.** Fibre optic for long-distance connections between buildings (handles 300m+ without degradation) and areas near electrical interference, copper for standard terminal connections where shorter distances and lower costs apply
- ○ **C.** Fibre optic cables cannot be used outdoors, so copper must be used everywhere

○ **D.** No cables are needed—modern stadiums should be entirely wireless

Q29 (3 marks)

A warehouse management system sends inventory updates to the company's headquarters across the country. Explain how TCP and IP work together to ensure the data arrives correctly.

○ **A.** TCP and IP are the same protocol with different names

○ **B.** IP handles reliability while TCP handles addressing—they have reversed roles

○ **C.** IP addresses and routes packets across networks to reach headquarters, while TCP ensures reliable delivery by managing packet sequencing, error checking, and retransmission of lost data

○ **D.** TCP and IP only work within a single building and cannot send data across the country

Q30 (3 marks)

An architecture firm with 15 employees currently uses standalone computers. They are considering implementing a network with centralised file storage. Evaluate the key advantages and disadvantages they should consider before making this decision.

○ **A.** There are no disadvantages to networking—the firm should implement it immediately without further analysis

○ **B.** Advantages include easier file sharing, centralised backup, and shared resources; disadvantages include setup costs, security requirements, and dependency on the server which could affect all users if it fails

○ **C.** Networking would prevent employees from saving any files, so they should remain standalone

○ **D.** The firm is too small to benefit from networking—networks only help organisations with more than 100 employees

Total: **55** marks • Time: **30 min**

↑ Back to top

Submit answers

Williams Physics Education — over **1,100,000+ marks** of exam-aligned content and **4,200+ auto-marked assessments** across 67 qualifications at GCSE, A-Level, AP & Professional Qualifications. Free to use, no login required.

About & Contact • Privacy • Terms • Accessibility • Copyright • DMCA • YouTube

WPE is free for everyone. If our resources help you or your students, you can support the project here.

© 2026 Williams Physics Education. All rights reserved.